

Mission Name

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Data Storage (DS)
Version 2.4.1.0

APPLICATION USER’S GUIDE

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Goddard Space Flight Center
Greenbelt, Maryland

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A

FORWORD

This Core Flight System (CFS) Data Storage (DS) Application User's Guide provides guidance for the Flight Operations Team (FOT) for the CFS DS Application.

This is one of a set of enhanced User's Guides for the CFS Product Documentation Suite. While the main audience is the FOT, the Guides also help serve the needs of flight software developers; Flight Software Sustaining Engineering (FSSE), Integration and Test (I&T), and others who support missions which use CFS.

This generic CFS DS Guide is set up so that missions can convert it into a mission-specific guide.

The signatures on this page apply to the document as distributed before mission to launch.

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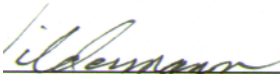
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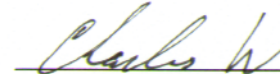


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C 1. Introduction to the CFS DS User's Guide

1.1 G

The primary purpose of this Application User's Guide is to help the Flight Operations Team (FOT) understand the DS application.

Many other purposes will be found for this Guide, including helping mission flight software personnel populate the ground system Record Definition Language (RDL) files in the ground system used later by the FOT, e.g., Advanced Spacecraft Integration & System Test software (ASIST).

Further purposes of this Guide are to help mission developers, system integration test team members, and FSSE to understand the DS application for their own specific needs, such as using the software to perform certain hardware tests.

If you want a more detailed understanding from a developer or FSSE standpoint, review separately the Doxygen-generated html documentation files, or review the source files directly.

As delivered, this is a generic document ready to insert mission defined values to serve the needs of specific missions.

1.2 A

This Application User's Guide relies heavily on the content of earlier heritage DS publications, presentations, and interviews with flight software engineers. Appendix A is based on information from CFS DS source code and reformatted for this publication. A publication like this is a team effort - thank you to the developers, the management team, and the entire Code 582 review team.

1.3 C

- In this document, *Data Storage Application*, *DS application*, and *DS* are used interchangeably, and refer to the CFS version unless specifically referring to heritage code.
- In this document, *flight controller*, *ground controller*, and *FOT* are used interchangeably.
- In this document, the percent sign (%) when followed by a string may indicate variable text. See Appendix A for references to the text that may be substituted in each case.
- *Core Flight Executive* is abbreviated cFE (lower case "c").

- *Partition* refers to the top level directory of a storage device. A storage device may be formatted so that it contains multiple partitions. Partitions may be fixed sized file systems or simply directory names.
- *File Location* refers to the full path file specification.
- *Pathname* refers to the full path file specification.
- *Basename* refers to the portion of the DS filename that follows the pathname, and precedes the sequence number.
- *Sequence number* refers to a counter, maintained by DS, which is used to uniquely identify a file. For example, filenames that use the size schema are named using the following notation: Basename + File Sequence Counter + Extension (e.g. InstrumentX12345678.xyz).
- *Full path file specification* refers to the Pathname, plus the Basename, plus a Sequence number, and finally, an extension. DS manages the “filename” as the full path file specification.
- *See Appendix B for more.*

1.4 **D**

Documents used in the preparation of this Guide are listed in the table below.

Table 1 Related Documents

	D	D	D
1	/A		, . <i>Core Flight System Data Storage (DS) Application Design As Built [Design Presentation]</i> . G , MD: G F C , C 582 (F B), 22 2001. .
2	/A		, . <i>CFS Data Storage (DS) User's Guide</i> . G : A A G F C , C 582, CF D , 1 M 2013. D . H ML.
3	582-2008-008		<i>CFS DS Requirements V1_1 080111</i> . G : A A G F C , C 582, F B , 1 A 2011. DF.
4	464-F - HDBK-0138		, . <i>Solar Dynamics Observatory (SDO) Telemetry Output (TO) Users Guide Version 2.0</i> . G : A A G F C , D F D , 18 A 2007. D C.

1.5 **A**

1.5.1

This

1.5.2 CF D A

The following list summarizes the assumptions made about the CFS DS Application as documented in this Guide:

- The CFS DS code has not been modified.
- The CFS DS Application has been configured using the standard CFS DS configuration parameters.
- The cFE Application Programming Interface (API), and Operating System Abstraction Layer (OSAL) are being used.
- Consultative Committee for Space Data Systems (CCSDS) File Delivery Protocol (CFDP) is being used.
- DS assumes that a file system is being used. In order to get the data to the ground, a file transfer mechanism such as CFDP must be used; DS is not responsible for transferring the data.
- DS does not provide the interface to the hardware used for storing the data. DS simply writes to files and assumes that the interfaces required to write to the devices/partitions are available.
- Use of Electrically Erasable Programmable Read Only Memory (EEPROM) generally requires that writing be enabled prior to writing. DS does not provide this capability.
- A command is sent to DS to request DS housekeeping messages (i.e. from the CFS Scheduling [SCH] application).
- Until a disk is created and partitions formatted for DS to use to store the destination file, the DS application should be disabled. Once the disk has been created, and partitions formatted, the Enable command can be sent to enable the DS application to start opening and writing to files, as defined in the Destination File Table.
- The command mnemonics shown in this Guide were developed for, and assume the use of, ASIST ground control software.

1.6 H D

Use this Application User's Guide as a learning tool before launch. Use the Appendix as a reference tool after launch. Read it offline as a paper printout, or search the electronic file. Use the Guide along with Doxygen generated .html files for more complete understanding. In contrast to this Guide, the Doxygen generated html files are primarily targeted to developers and FSSE.

Experienced flight controllers may only need to browse the Guide, and use the Appendix as needed. New flight controllers may wish to get more familiar with the entire Guide well before needing to rehearse operational scenarios or set up flight software procedures ("procs").

1.6.1 F

For CM reasons, if you find any item in this Guide that is in error, or want to be informed of any updates, please email feedback to the cFE/CFS PDL for validation and routing. As of the date of publication, the cFE/CFS PDL is Susie Strege (susanne.L.strege@nasa.gov).

Besides corrections of any errors, the CFS team is interested in your ideas on improvements that help understanding.

1.7 A A

Acronyms and abbreviations in this publication are shown in Table 2 below. Telemetry, command mnemonics, and similar terms are not shown here.

Table 2 Acronyms and Abbreviations

A A	D
AC	A C
A I	A I
A ID	A ID, M ID
A I	A I &
C&DH	C D H
CC D	C C D
CC	C C
CD	C D
CFD	CC D F D
FE/CFE	C F E
CF	C F
CI	C I
C	C
D	CF D
EE M	E E M
F	F
F E	F E
G &C	G C
G M	G M
G FC	G F C
HK	H A
I&	I
I	I
L	L
L B	L B
MID	M ID

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C 2. Introduction to the CFS DS Application

This chapter addresses the theory and internal operation of the CFS DS application.

2.1 H

The CFS DS code derives from the Lunar Reconnaissance Orbiter (LRO) mission, which used data storage files. Most other heritage missions, such as the Solar Dynamics Observatory (SDO) and Space Technology 5 (ST-5), stored the data as messages or Virtual Channel Data Units (VCDUs) in buffers. In the CFS DS design, data is stored as cFE Software Bus messages in files, rather than buffers.

The CFS team took LRO's DS application and made it CFS compliant, adding configuration parameters in place of hard coded parameters, and making other changes to fit it into the larger CFS architecture.

2.2 CF D

The CFS DS application performs three basic functions:

- It writes table-defined data packets (cFE Software Bus messages), into files for later transmission to the ground.
- It performs filtering on received packets, prior to storage.
- It stores files on a file system.

DS references two tables, the Packet Filter Table and the Destination File Table.

2.2.1 -D D F

DS writes cFE Software Bus messages to table-defined data storage files. These files are generally stored on a device such as a solid state recorder. However, they could be stored on any storage device with a file system.

DS has no file download or playback capabilities; another application such as CFDP must be used to transfer to the ground the files created by DS.

2.2.2 D

Any cFE Software Bus message (a CCSDS compliant packet) can be written to files by DS.

- The cFE Software Bus (SB) application routes cFE Software Bus messages to DS. The messages can come from a number of sources; these sources can include ground commands via a command ingest application.

- The cFE Table Services application provides DS with knowledge of updates to the DS Destination File Table and the DS Packet Filter Table.

DS keeps a list of Message IDs (also known as Application ID, or *ApID* entries) to which it has subscribed. For fast look up of packet ApIDs, DS creates a hash table.

For the Packet Filter Table (though not the Destination File Table) when DS receives notice from cFE Table Services that a new table commit is pending, DS unsubscribes to the ApID entries in the old Packet Filter Table before the new table data is copied into DS table space. Typically the table service command sequence is load/validate/commit. DS then subscribes to the new ApIDs and creates a new hash table.

2.2.3 F D

DS can store cFE Software Bus messages at different rates using two possible filtering schemes (Sequence or Time Based filtering) allowing "N of X" messages to be stored starting at offset 0 (zero).

2.2.3.1 F

Each Packet Filter Table is identified by a character string descriptor whose length is configurable by the mission. Typically, a mission may set up multiple versions of the Packet Filter Table, such as for I&T; launch; orbit insertion; Guidance and Navigation; Attitude Control System (ACS); and nominal, for example.

The Packet Filter Table contains a *list of packets* (Message IDs) that DS will process, along with *filtering information* specific to each packet.

For each Message ID, the Packet Filter Table specifies the *storage and filtering information* shown in Figure 1 below.

Storage Element

- A zero based index into the Destination File Table

Filtering Elements

- The type of filter used by this entry: by count or by time (the filter type defines whether the input data is a value derived from the packet timestamp, or if the input data is derived from the header sequence count)
- The N and X arguments passed into the Filter algorithm that allows “N of every X” data packets to be stored
- The O argument passed into the Filter algorithm Filter that allows data packets to be stored starting with the packet at offset “O”

Figure 1 Description of Entry in the Packet Filter Table

The configuration parameter DS_FILTERS_PER_PACKET specifies the **storage and filtering information** per packet. This allows a packet to be stored in multiple destinations at different rates.

For example, each full entry would contain a Message ID, accompanied by:

$[(\text{Storage Element, Filtering Elements})^1, (\text{Storage Element, Filtering Elements})^2, \dots (\text{Storage Element, Filtering Elements})^A]$ where A is the number specified by the configuration parameter.

The **filtering elements** include three (3) 16-bit parameters (N, X, and O). The filter algorithm allows “N of X” cFE Software Bus messages to be stored starting at offset O.

2.2.3.2 F

Filtering by sequence uses the packet sequence number to determine if the cFE Software Bus message should be filtered or not.

The filter factor has three parameters and can be expressed in terms of N number of packets out of X number of packets. The offset Y determines the starting point within X for selecting packets.

If X is greater than zero, the message will be stored if and only if the sequence number modulo X is less than N. For example, N=4, X=6 and O=2 will store 4 of 6 messages starting at offset 2.

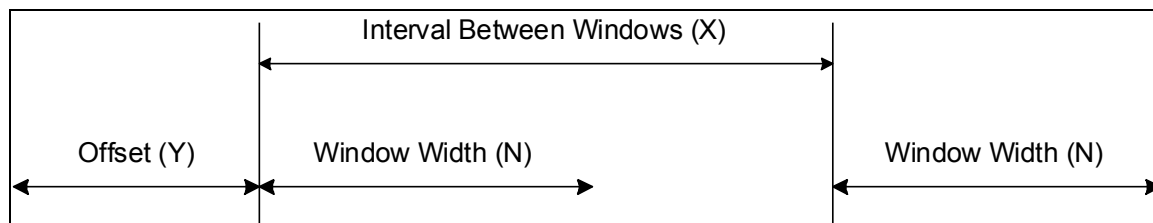


Figure 2 Filter Factor Parameters - Filtering by Sequence

If the packet sequence number minus Y mod X is less than N then the packet will be output:

```

if ((( acke _ e e ce_c e - Y) MOD (X)) < N) he
    he acke
el e
    d he acke
e d if

```

2.2.3.2.1 Filtering by Sequence, Example 1

In Filtering by Sequence - Example 1, we send two out of every four packets, offset packet zero. This will send two packets and then skip two packets. If instead one wanted every other packet then one would specify one out of two packets.

Table 3 Filtering by Sequence – Example 1

N	2
Y	0
X	4

Sequence Number	Send
0	✓
1	✓
2	
3	
4	✓
5	✓

Sequence Number	Send
6	
7	
8	✓
9	✓
10	

2.2.3.2.2 Filtering by Sequence, Example 2

In Filtering by Sequence - Example 2 we send two out of every four packets, offset packet one. This will send two packets and then skip two packets, starting with packet one:

Table 4 Filtering by Sequence – Example 2

N	2
Y	1
X	4

Sequence Number	Send
0	
1	✓
2	✓
3	
4	
5	✓
6	✓
7	
8	
9	✓
10	✓

2.2.3.3 F

Time based filtering uses the Spacecraft time code found in the CCSDS packet secondary header. The filter factor has three parameters and can be expressed in terms of N number of packets out of X number of Seconds/Sub Seconds. Note there are 16 Sub Seconds per Second.

The offset Y determines the starting point within the X for selecting packets. Messages are stored if and only if the message time stamp modulo X is less than N.

Fifteen bits out of the 64-bit CCSDS time code field are used. The constructed 15-bit time stamp used for DS filtering consists of the 11 least significant bits (LSBs) of the seconds field and the four most significant bits (MSBs) of the sub-seconds field. Messages will be stored if and only if the constructed 15-bit time stamp modulo X is less than N.

This type of filtering is rare in that the N value must be set to the desired 15-bit time stamp, which is hard to specify.

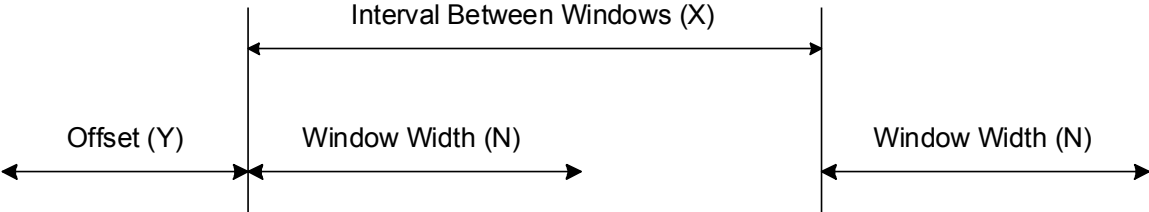


Figure 3 Filter Factor Parameters - Filtering by Spacecraft Time

The time code filter factor consists of the least significant 11 bits of the Lo_Word Seconds (a), and the most significant four bits of the Hi_Word Sub Seconds (b). Bit 16 is not used (c). For more, see tables below.

Table 5 Filtering by Time – Time Code Format from Packet

Hi_Word Seconds	Lo_Word Seconds	Hi_Word Sub Seconds	Lo_Word Sub Seconds
	Xxxxxaaaaaaaaaaa	bbbbxxxxxxxxxxxxx	

Table 6 Filtering by Time – Time Code Filter Format

c	aaaaaaaaaaa	Bbbb
---	-------------	------

The sign of X determines the type of filtering that will be performed:

```
if (X = 0)  he
    d '      a  acke
el e if (X > 0)  he
    e e ce    be fil e i g
el e if (X < 0)  he
    i e ba ed fil e i g
e d if
```

If the packet time code (in time code filter format) minus Y mod X (the absolute value in time code filter format) is less than N then the packet will be output.

```
if ((( acke _ i e _ c de - Y) MOD (ABS(X))) < N)  he
    he acke
el e
    d      he acke
e d if
```

2.2.3.3.1 Filtering by Time, Example 1

In Filtering by Time - Example 1, if a packet is generated four times a second, we send one packet per second. An offset of zero (0) will send the first packet generated per second. This works well for synchronous data that is generated at exactly the same time each second.

Table 7 Filtering by Time, Example 1

N	1
Y	0
X	16

Time (Sub Seconds)	Pkt Gen	Send
0	Pkt	✓
1		
2		
3		
4	Pkt	
5		
6		
7		
8	Pkt	
9		
10		
11		
12	Pkt	
13		
14		
15		
16	Pkt	✓

2.2.3.3.2 Filtering by Time, Example 2

In Filtering by Time - Example 2, no packets would be sent because no packets are generated in time slot zero (0). Note that the offset must be used to synchronize the time slot that we want to send with the time slot in which the packet is generated. The offset must be set to one (1) to make the filtering work the same as in the previous example.

Table 8 Filtering by Time, Example 2

N	1
Y	0
X	16

Time (Sub Seconds)	Pkt Gen	Send
0		
1	Pkt	
2		
3		
4		
5	Pkt	
6		
7		

Time (Sub Seconds)	Pkt Gen	Send
8		
9	Pkt	
10		
11		
12		
13	Pkt	
14		
15		
16		
17	Pkt	

2.2.3.3.3 *Filtering by Time, Example 3*

In Filtering by Time - Example 3, if a packet is generated four times a second, we send one packet per second. An offset of four (4) will send the second packet generated per second. This works well for synchronous data that is generated at exactly the same time each second. Note that an offset of one, two, or three (1, 2, or 3) will not send any packets because packets are not generated in those time slots.

Table 9 Filtering by Time, Example 3

N	1
Y	4
X	16

Time (Sub Seconds)	Pkt Gen	Send
0	Pkt	
1		
2		
3		
4	Pkt	X
5		
6		
7		
8	Pkt	
9		
10		
11		
12	Pkt	
13		
14		
15		

2.2.3.3.4 *Filtering by Time, Example 4*

In Filtering by Time - Example 4, if a packet is generated 10 times a second, we send two packets per second. Note that the X is 1/2 a second or eight sub seconds.

Table 10 Filtering by Time, Example 4

N	1
Y	0
X	8

Time (Sub Seconds)	Pkt Gen	Send
0	Pkt	X
1		
2	Pkt	
3		
4	Pkt	
5	Pkt	
6		
7	Pkt	
8	Pkt	X
9		
10	Pkt	
11		
12	Pkt	
13	Pkt	
14		
15	Pkt	

2.2.3.3.5 *Filtering by Time, Example 5*

In Filtering by Time - Example 5, if a packet is generated four times a second, we send two packets per second. Note that in this example packets are not generated at exactly the same times each second and we will not see every packet we expect.

Table 11 Filtering by Time, Example 5

N	1
Y	0
X	8

Time (Sub Seconds)	Pkt Gen	Send
0		
1	Pkt	
2		
3		
4	Pkt	
5		
6		
7		
8	Pkt	X
9		
10		
11		
12		
13	Pkt	

14		
15		

2.2.3.3.6 *Filtering by Time, Example 6*

To fix the problem in example 5, and to get every other packet, set the N to 3. This will send the packet as long as it falls within 3 slots (3 packets every 8 sub seconds).

Table 12 Filtering by Time, Example 6

N	3
Y	0
X	8

Time (Sub Seconds)	Pkt Gen	Send
0		
1	Pkt	X
2		X
3		
4	Pkt	
5		
6		
7		
8	Pkt	X
9		
10		
11		
12		
13	Pkt	
14		
15		

2.2.4 D F

The Destination File Table defines a list of data storage files along with creation information specific to each file. A configuration parameter determines the maximum number of files that can be defined in the Destination File Table.

The Destination File Table defines data storage files as shown below:

Filename

- The file path
- The base portion of the filename
- The filename extension (including the dot)
- The file type: by count or by time (the file type denotes whether the file name includes an incrementing value or the Spacecraft time)

State

- The enable/disable state for this destination file

Limits

The controlled copy of this document is located online at <https://fsb.gsfc.nasa.gov/CFS/>

- The file **size** at which to close the file and open a new one
- The file **age** at which to close the file and open a new one

Sequence

- An initial value for a numerical sequence to be appended to the filename before the extension

As each file is closed, if the file type is by count, then the file type counter is incremented for the next file to be opened. (If the file type is by time, then the current Spacecraft time is always used, and Spacecraft time is always changing, resulting in a unique filename for each file.)

Each DS filename contains characters derived from a filename sequence counter or from the current Spacecraft time at the time the file was created.

Files are generally organized in a way that is meaningful to each mission. For example, some missions may wish to include all housekeeping data into one set of files, while others may wish to split out certain types of housekeeping data into separate files. The Destination File Table contains the scheme, or definitions, for the files into which cFE Software Bus messages will be saved.

Any packet sent on the Software Bus can be stored in a file. Typically Command and Data Handling (C&DH) housekeeping telemetry is saved into one file, for example, and Guidance, Navigation and Control (GN&C) housekeeping in another file. Instrument data may be saved in yet another file.

Science data, if it is low volume, may also be saved, though typically science data is larger volume and handled by a separate, dedicated module.

The Destination File Table is typically set up before launch and is not frequently changed.

2.2.4.1 D F

DS manages the filename as the Pathname, plus the Basename, plus a Sequence number, and finally, an extension. The entire length of the filename must not exceed the length specified by the OSAL configuration parameter OS_MAX_PATH_LEN.

Each entry in the Destination File Table contains a character string definition for Pathname, Basename and Extension and a type selection for the Sequence portion of the filename.

Depending on the Filename Type, DS will construct a character string that represents either the value of a filename sequence counter specific to that particular Destination File Table entry, or the current time of day in the format YYYYDDHHMMSS.

If the file is sequence-based, the sequence characters will be based upon the next sequence number. For example, if the initial value for a numerical sequence = 1, Pathname = /ram and Basename = seq and extension = .dat, then the file would be named /ram/seq00000100.dat for the 100th file.

- For the maximum value a file sequence count will reach before rollover to zero, refer to the mission-specific DS config. parameter DS_MAX_SEQUENCE_COUNT.
- For the number of filename sequence count digits used when the filename type has been set to "count" rather than "time," refer to the mission-specific DS configuration parameter DS_SEQUENCE_DIGITS.

Tip: As a reminder, the only portion of **sequence-based** filenames that change is the sequence counter, as assigned to the file at the time the file is opened. The path, base, and extension are

constant as defined in the Destination File Table, unless changed by command and/or a table load.

Tip: Note that in sequence-based filenames the sequence count always rolls over to zero. The rollover to zero is not configurable to a different number. In practice, missions often reset the sequence count each orbit, or whenever the mission changes the base part of the filenames to identify a new data set. So the result is that in practice the numbers seldom get large enough to roll over.

If the file is **time-based**, the characters will be in the form YYYYDDDDHHMMSS. For example if the file was created on day 123 at 14:15:22 using Pathname = /ram, Basename = tim and extension = .dat, the file would be named /ram/tim2010123141522.dat.

Tip: As a reminder, the only portion of **time-based** filenames that change is the Spacecraft time, as assigned to the file at the time the file is opened. The path, base, and extension are constant as defined in the Destination File Table, unless changed by command and/or a table load.

For the path, base and extension, refer to the mission-specific File Table for the appropriate entry.

2.2.4.2 F

Each DS destination file contains a primary cFE file header with the information shown below.

Table 13 Primary File Header Contents

E	D
Content Type	Specifies cFE compliant files. Value will be set to 0x63464531
SubType	Specifies type of file. Value is defined via DS configuration parameter
Primary header length	Length of primary header
Spacecraft ID	Spacecraft that generated the file
Processor ID	Processor that generated the file
Application ID	Application that generated the file
File creation time	In seconds
File creation time	In sub-seconds

2.2.4.3 F

Each DS destination file contains a mission-optional secondary file header. Each mission has the option (controlled by configuration parameter) to choose not to include a secondary file header.

The format of the secondary file header is also configurable. This version of the DS application includes the ability to generate the CFS DS standard file header or the file header that conforms to the specifications of the Global Precipitation Measurement (GPM) mission. The tables below shows the contents of the DS standard file header and GPM file header.

Table 14 Secondary File Header Contents

E	D
File close time	Seconds portion of Spacecraft time that the file was closed
File close time	Subseconds portion of Spacecraft time that the file was closed
File table index	Destination File Table index
File name type	File type (count vs. time)
Qualified filename	Full path specification of file

Table 15 GPM File Header Contents

E	D
Source ID	String of four characters identifying the source of the file data
Sequence ID	File generation counter
File start time	Seconds portion of Spacecraft time that the file was opened
File stop time	Seconds portion of Spacecraft time that the file was closed
File size	Total size of the file in bytes
Spare	Spare bytes (20 total)

2.2.4.4**C**

.

For configuration management, some missions prefer to make changes only by changing tables via a table load where a new table file can be uplinked via CFDP and loaded via cFE Table Services. However, DS supports changing individual table entry settings by command. For details, see Appendix A, section A.4, DS Commands.

2.2.4.5 C**F**

Files can be defined to close at a maximum size (measured in bytes) and a maximum age (measured in seconds). When closed, a new file will be opened.

If the desired **maximum file size** would be exceeded, the application will close the file and open a new one with a unique name according to the rules set up in the Destination File Table. If the desired **maximum age** would be exceeded, the application will close the file and open a new one with a unique name according to the rules set up in the Destination File Table.

However, files are not closed based on whether the filename is sequence based or time based. **Both the file size and file age are checked.** If a file is time based but the file size is exceeded before the time is met, the file will still be closed.

If one wants the file to close by *age* then the file size limit needs to be set unreasonably large. The unreasonably large file size limit should never pass, making age the only meaningful check.

Similarly, if one wants the file to close by *file size* then the age limit needs to be set unreasonably large. The unreasonably large age limit should never pass, making the file size the only meaningful check.

To avoid creating empty files, the desired file size is only verified after the first packet is saved. So the first packet written may cause the file to exceed the desired size.

DS will always have at least one packet in every file. Its size is defined in the Destination File table, and can be changed by the FOT.

cFE Software Bus messages are stored in a file or files in their entirety, i.e. messages are not segmented between files.

2.3 **D** **C**

The figure below shows the software context for the CFS DS Application.

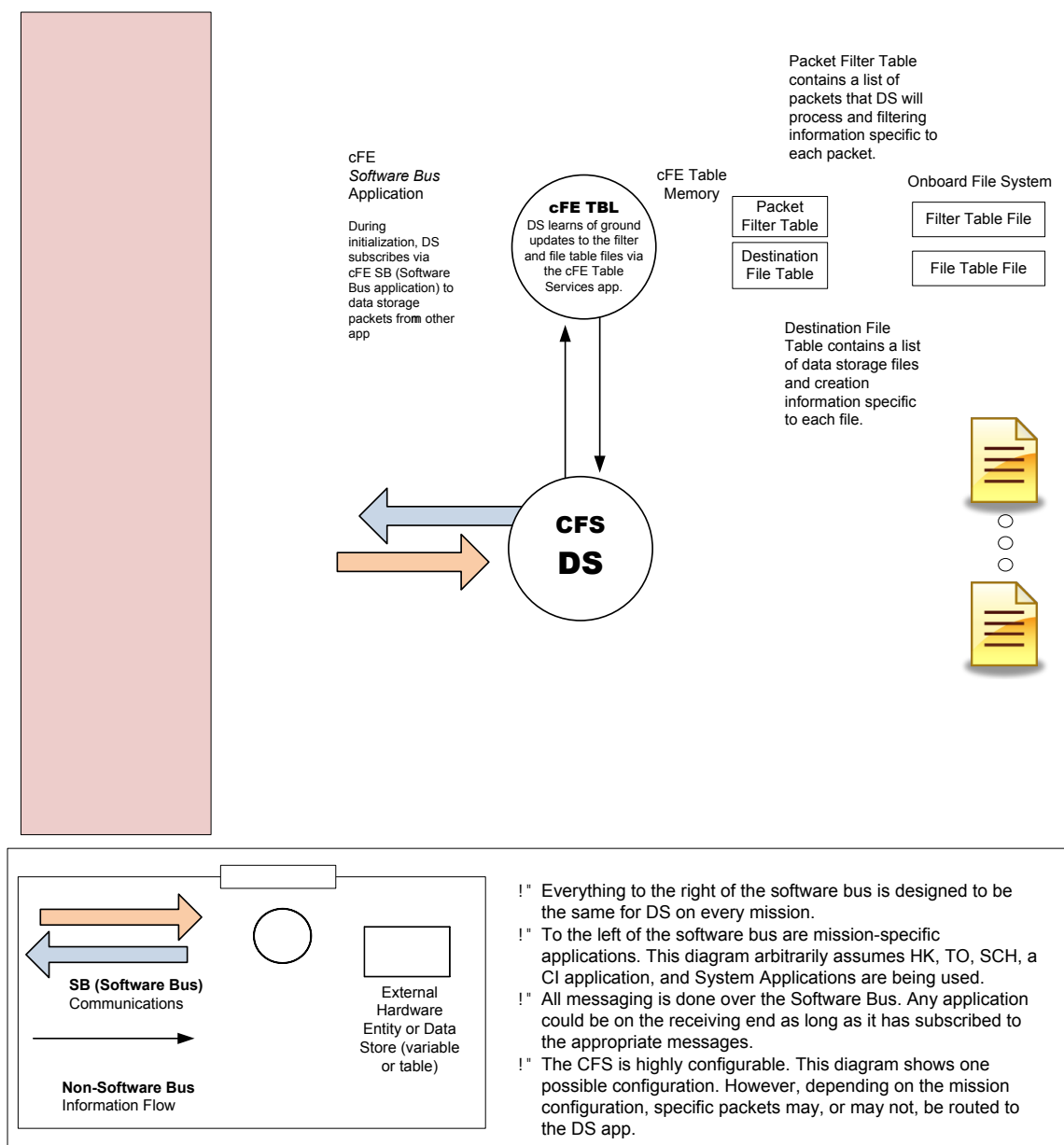


Figure 4 DS Application Software Context

2.3.1 A

As shown in Figure 4 above, CFS system applications as configured for a particular mission may communicate with DS via message packets via the cFE Software Bus application.

2.3.2 C A

Ground commands are routed through a CI Application, as shown in Figure 4 above.

2.3.3 FE B A

Message packets which will be filtered and saved by DS arrive via the cFE SB Application. As shown in Figure 4 above, CFS applications communicate with DS via SB.

2.3.4 C A

The SCH application wakes up DS when SCH requests DS housekeeping data, as shown in Figure 4 above. Additionally, DS subscribes to message packets, and is awakened when it receives those packets.

2.3.5 FE (B) A

As shown in Figure 4 above, DS learns of any updates to the DS Packet Filter Table and Destination File Table through the cFE Table Services (TBL) application.

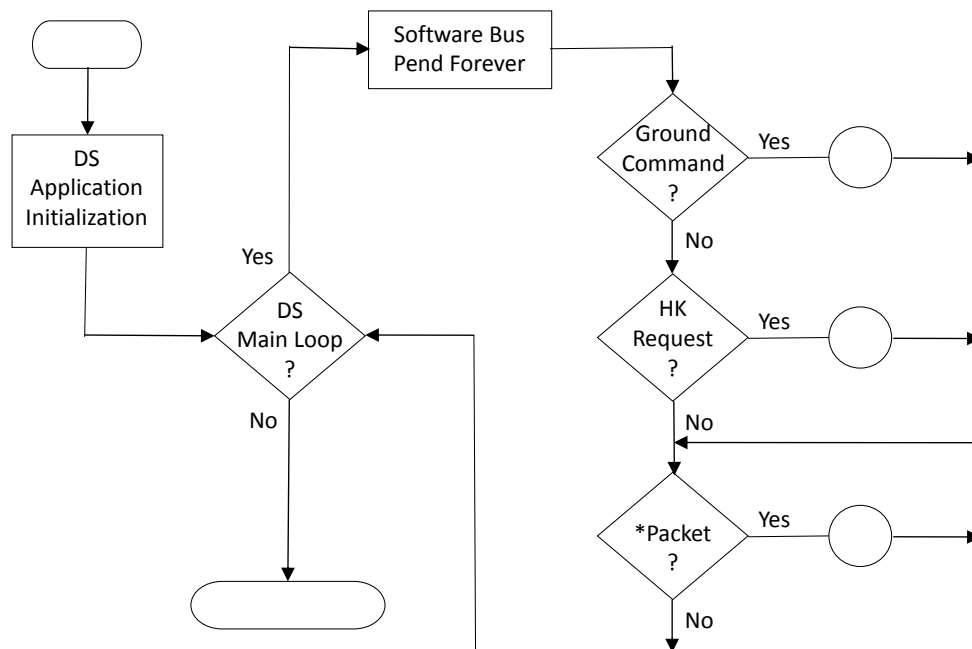
2.3.6 FE (E)

cFE TIME (the cFE Time application) distributes Spacecraft time of day data used by DS, as shown in Figure 4 above.

2.3.7 (), C (C)

As shown in Figure 4 above, any cFE Software Bus messages generated by DS are routed to whatever mission-specific applications subscribe to them, such as the Housekeeping (HK) application, a telemetry output (TO) application, and/or Stored Command (SC) Applications.

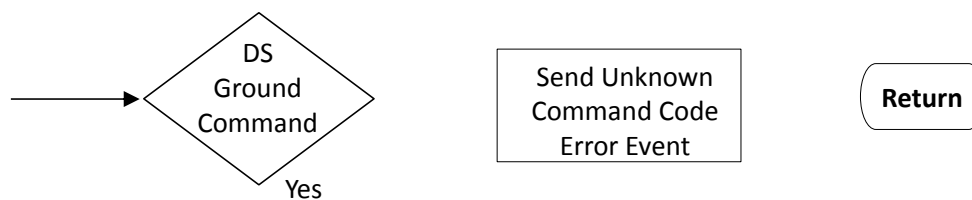
Figure 5 below shows the overall program flow of the DS, from start to exit.



*A Software Bus Packet (Message ID) that DS has subscribed to

Figure 5 DS Overall Flow Control

Figure 6 below shows the flow control of DS for *ground* commands.



- 0 - No-op (display version)
- 1 - Reset Housekeeping Counters
- 2 - Set DS Application State
- 3 - Set Packet Filter File Index
- 4 - Set Packet Filter Type
- 5 - Set Packet Filter Parameters
- 6 - Set Destination File Type
- 7 - Set Destination File State
- 8 - Set Destination File Pathname
- 9 - Set Destination File Basename
- 10 - Set Destination File Extension
- 11 - Set Destination File Max Size
- 12 - Set Destination File Max Age
- 13 -

Figure 6 DS Flow Control Detail (A) - Ground Command

Figure 7 below shows the flow control of DS for *housekeeping* requests.

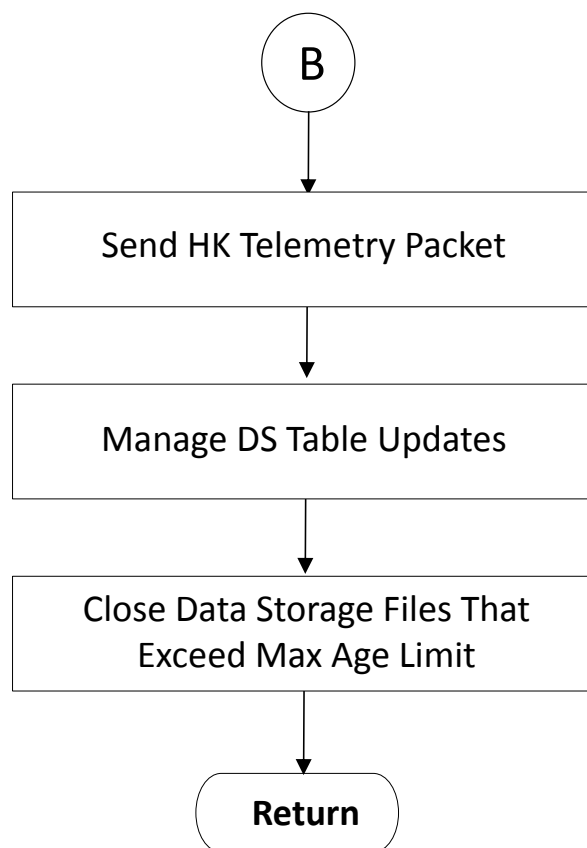


Figure 7 DS Flow Control Detail (B) - Housekeeping Request

Figure 8 below shows the flow control of DS for *Store Data*.

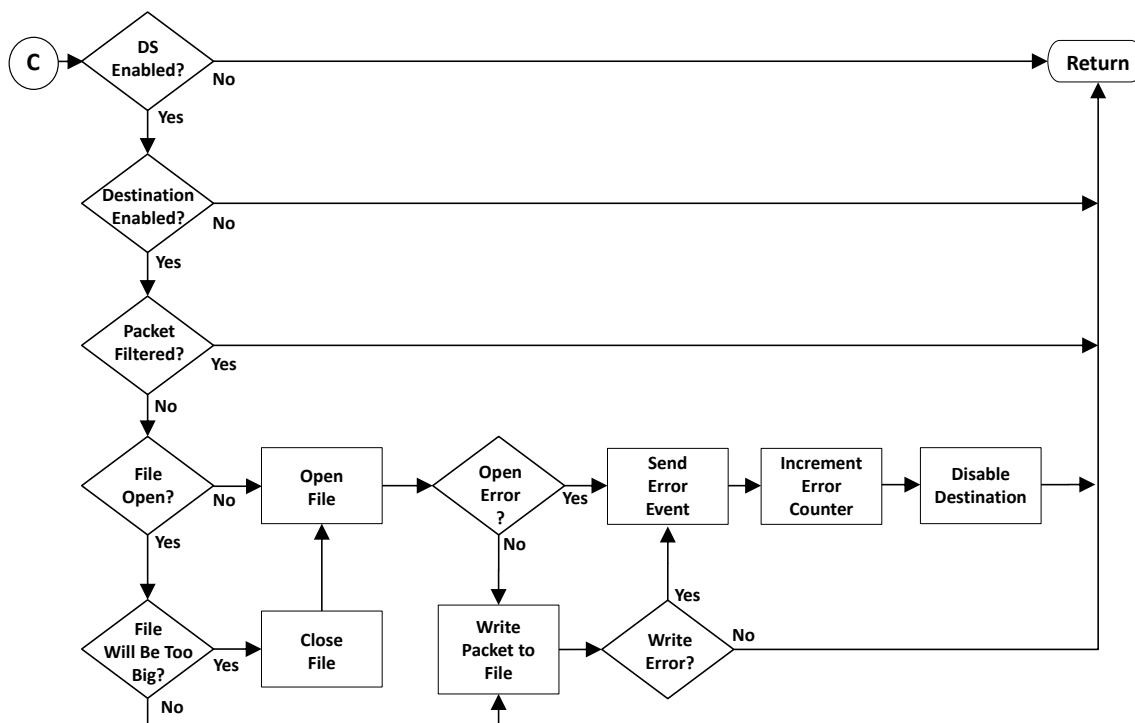


Figure 8 DS Flow Control Detail (C) - Store Data

Figure 9 below shows a visual description of how the two DS tables are used to control data storage.

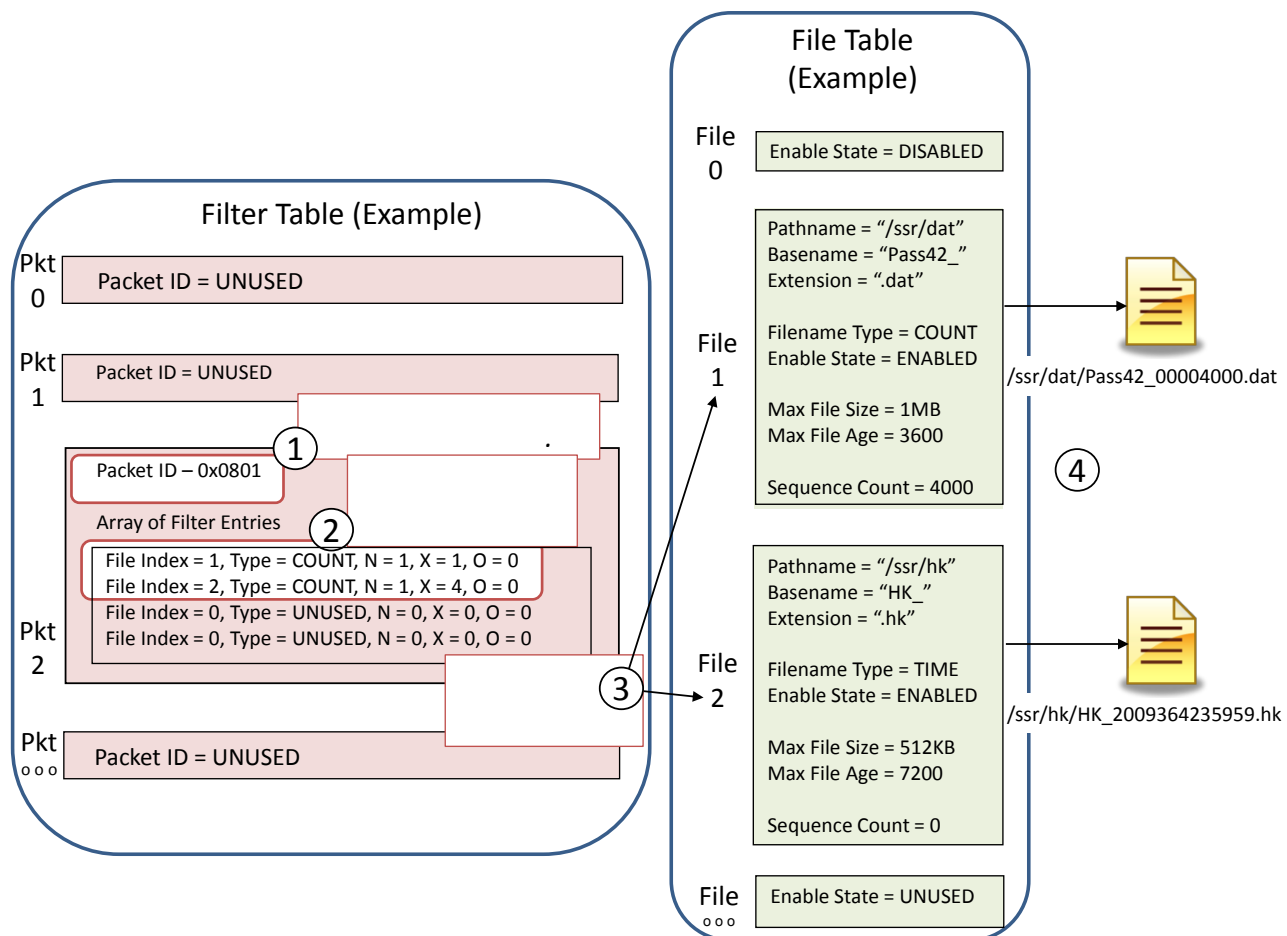


Figure 9 DS Input Packet Flow (Example)

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C 3. CFS DS Normal Operations

This chapter addresses the most common issues expected to arise during normal operations.

3.1 D

DS can be configured to start up enabled or disabled. A configuration parameter (DS_DEF_ENABLE_STATE) defines the state in which DS will start following a power-on reset. Another configuration parameter (DS_CDS_ENABLE_STATE) allows DS to restart in the last commanded state or DS_DEF_ENABLE_STATE following a processor reset.

In **enabled mode**, DS will begin to process packets immediately on startup. Processing packets means DS is checking to see if the received CFS Software Bus Message is included in the Packet Filter Table. If the message is included in the Packet Filter Table, DS runs the filtering algorithm to determine if the packet should be written to a file or not.

Upon startup or table update, if the Destination File Table and/or Packet Filter Table have failed to load, DS will not store any messages until both tables are loaded with valid data. Loading the Tables with valid data can be accomplished by using the cFE Table Services commands after creating a Table load image file and uploading it to the spacecraft's file system. Until both tables are loaded with valid data, DS will increment the IgnoredPktCounter for each received packet that is discarded.

The flow of this process is shown starting in Figure 8, DS Flow Control Detail (C) - Store Data, and described below starting in Section 3.2, DS Operation.

In **disabled mode**, DS will ignore message packets contained in the Packet Filter Table that DS has subscribed to until receipt of a valid command to enable the DS Application. DS will not increment the IgnoredPktCounter for each received packet that is discarded when in **disabled mode**.

3.2 D

When DS is enabled for storage and DS tables have loaded successfully, DS works as follows:

1. When DS receives a cFE Software Bus message that passes the filtering algorithm, the message is stored in a file or files based upon the Packet Filter Table information for that Message ID. The Packet Filter Table points to the index or indexes of the file or files defined in the Destination File Table. Once a message is stored, the DS housekeeping packet provides information about the open file.
2. Each housekeeping cycle, DS evaluates open files to determine if their maximum age has been exceeded. If their maximum age has been exceeded, then DS closes the file. The age test is done at housekeeping time because an old file needs to be closed even if there are

no packets being received. Effectively, this means the age is tested about every five seconds or so.

3. The check for maximum size is done as DS receives cFE Software Bus messages. If DS and the destination file are enabled, the destination file is open, and the cFE Software Bus message passes any filtering, then the size check is performed. If the total of the destination file's current size plus the new cFE Software Bus message size exceeds the file size limit, then the file is closed. DS opens a new file, and writes the cFE Software Bus message to the new file. A minimum of one message is stored in each file.

3.2.1 F

The Packet Filter Table determines what cFE Software Bus messages DS will store via the Software Bus Message IDs and the rate the incoming messages will be stored. The DS application supports two types of filtering: *sequence number based* filtering and *time based* filtering.

Each filtering type relies on three filter factors, N, X, and O where N of X messages will be stored starting at offset O.

DS may store a single cFE Software Bus message using different filter schemes.

For more detail on filtering refer to Chapter 2.

3.2.2 D

D A B

The Message IDs to be stored are specified in the Packet Filter Table. The currently loaded Packet Filter Table may be dumped via cFE Table Services to obtain a listing of the stored Message IDs. Once the table is dumped and downlinked, the table contents are typically displayed on the mission-specific Packet Filter Table display page in ASIST, Integration Test and Operations System (ITOS), or other ground flight software.

3.2.3 D

F A B

The locations of the files are determined by both the Packet Filter Table and the Destination File Table. The Packet Filter Table, for each Message ID, indicates the index(s) into the Destination File Table, specifying the file(s) in which the message is to be stored. For details about information in the Packet Filter Table, see Section 2.2.3.1, Packet Filter Table.

The Destination File Table defines data storage files and their location. For details about how the Destination File Table defines data storage files, see Section 2.2.4, Writing Destination Files.

To trace the Message ID to the File entry, both tables must be dumped and downlinked. Once the tables are dumped and downlinked, the table contents are typically displayed on the mission-specific Packet Filter Table display page in ASIST, ITOS, or other ground flight software.

3.3 D

During initialization, DS subscribes to cFE Software Bus messages from other applications as defined in the Packet Filter Table.

DS attempts to load the Destination File Table and Packet Filter Table defined in the configuration parameters DS_DEF_DEST_FILENAME and DS_DEF_FILTER_FILENAME upon startup.

If the load fails for either of these tables, DS will still start. However, DS will not store any messages until both tables are loaded with valid data. This can be accomplished by using cFE

Table Services commands after creating a table load image file and uploading it to the spacecraft's file system.

3.3.1 -

On power-on reset, DS performs a cFE application-specific initialization:

1. Initializes housekeeping data and global variables.
2. Registers for cFE Services.
3. Creates Software Bus input pipe.
4. Subscribes to ground commands and housekeeping requests.
5. Creates Packet Filter Table and Destination File Table.
6. Loads table data from default table files.
7. Subscribes to message packets named in the Packet Filter Table.

3.3.2

On processor reset, all files that were open before processor reset will be closed, regardless of file size or age.

On processor reset, DS performs the same initialization as power-on reset unless the mission has implemented Critical Data Store (CDS), as described below.

1. If the configuration parameter DS_MAKE_TABLES_CRITICAL is set to "1", then cFE Table Services will load data for both DS tables from the CDS instead of from the default table data files.
2. Regardless of table criticality, DS will at startup attempt to retrieve the current filename sequence counters and destination enable/disable state values for each entry in the Destination File Table that is also stored in the CDS.

3.4 **E**

There are four levels of event types:

- Critical
- Error
- Information
- Debug

The levels are not used for programmatic control by CFS or DS. In other words, CFS and DS do not treat the level labels differently. However, ASIST, ITOS, or other ground flight control software may display event messages in different colors or otherwise differentiate them.

Debug level messages are typically used before launch for testing but then reconfigured off to avoid cluttering flight operator telemetry.

3.5 **D**

DS files are not created until a packet has passed the filter tests and is ready to be written to the file. This avoids potentially empty files.

Each packet received by DS (including DS commands and the DS housekeeping packet) is processed as a candidate for being written to a file.

If DS is enabled for packet processing, and the Message ID is found in the Packet Filter Table, then the packet filters are processed by evaluating the filter algorithm using parameters specific to each packet filter.

- If any passed packet filter references an enabled entry in the Destination File Table then the packet will be written to that file. A single packet may thus be written to multiple DS files.
- If the DS file does not yet exist, it will be created and initialized with both a cFE file header and a DS secondary file header.

3.6 **F D**

The Packet Filter Table consists of a series of packet entries. Each packet entry contains the

	at offset “O of X”.)
Algorithm_O	The O argument passed into the Filter algorithm

3.7 **D** **F** **D**

The Destination File Table contains the entries described below.

Table 17 Destination File Table Contents

E	D
Pathname	The full path file specification. Also known as file location.
Basename	Contains the base portion of the filename.
Extension	Contains the extension (including the dot).
Filename Type	Defines the sequence portion of the filename as either an incrementing value (“count”) or time of day.
File State	Defines the enable/disable state for this destination file.
Max File Size	Close current file when it gets this big.
Max File Age	Close current file when it gets this old.
Sequence Counter	Initial value for filename sequence counter. This value is referenced only when filename type is “count”.

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C 4. **Understanding DS Operational Constraints**

4.1 **F C**

4.1.1 F A

Each entry in the Destination File Table contains a maximum age value. This value represents the length of time, in seconds, that a particular data storage file may remain open.

As part of processing the housekeeping request command, DS checks the age of all open data storage files.

If the current file age exceeds the maximum age, the DS will close the file. A new file will be created only when another packet is received that passes the filter tests and targets that particular entry in the Destination File Table.

4.1.2 F

Each entry in the Destination File Table contains a maximum file size value. This number represents the maximum file size, in bytes, for a particular data storage file.

Prior to writing a packet to an existing data storage file, DS performs a test to see if the current file size plus the size of the packet would exceed the maximum size for that particular file.

If the resulting file size would exceed the maximum file size, then DS will close the old file and open a new file before writing the packet data.

Note that there is no size test performed on the first packet written to a new data storage file. This scheme ensures that every packet can be written to a file and that there are no empty files, but at the cost that a file with a single large packet may exceed the size limit.

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C 5. Frequently Asked Questions (FAQs)

5.1 H D ?

DS has nothing to do with downlinking packets and files. A file delivery application such as the CFDP (CF) application is required to downlink the files created by DS.

5.2 A ?

One early task for FOT is to set up the Packet Filter Table to limit collection of data to the desired amount between downlinking times. To do this, FOT should understand the Packet Filter Table. Remember that the Destination File Table doesn't have anything to do with limiting data. The Packet Filter Table alone "limits" the data.

5.3 H ?

There may arise a need to record all data when either not in contact with the ground, or when in contact, if the downlink rate is not high enough for all data. Later, at some point, it can be decided whether to downlink or delete the files.

There are no settings to keep files open indefinitely, but to effectively bypass size and age settings, these can be defined at the maximum value (4 billion hex), a value higher than either disc size or mission lifetime, respectively. The application will continue to check the values, but this setting will effectively keep the file open until manually closed by command.

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Table 21 Telemetry Data - Destination Table Error Counter

[illegible]

D	16 B I	16 B I
()	<i>CFS DS Default: \$sc_\$cpua_DS_FileUpdCnt</i>	

Table 25 Telemetry Data – File Update Error Counter

	F	E C
D	F	E C contains the count of the n
D	16 B I	
()	<i>CFS DS Default: \$sc_\$cpu_DSFileUpdErrCnt</i>	

Table 26 Telemetry Data – File Write Counter

	F	C
D	F	C
D	16 B I	
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileWriteCnt</i>	

Table 27 Telemetry Data - Total Number of File Write Errors

	F	E C
D	F	E C
D	16 B I	
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileWriteErrCnt</i>	

Table 28 Telemetry Data - Filtered Packet Counter

	F	C
--	---	---

D	F C contains a count of packets discarded, because they failed all filter tests, including
D	32 B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FilteredPktCnt</i>

Table 29 Telemetry Data – Packet Filter Table Error Counter

	F E C
D	F E C FE F .
D	E B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FilterPtrErrCnt</i>

Table 30 Telemetry Data – Packet Filter Table Load Counter

	F L C
D	F L C FE F
D	E B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FilterLoadCnt</i>

Table 31 Telemetry Data - Count of Packets Discarded - Packet Has No Filter

	I C
D	I C

()	CFS DS Default: \$sc_\$cpu_DS_IgnoredPktCnt
-----	---

Table 32 Telemetry Data - Passed Packet Counter

	C
D	p C contains the count of the number of .
D	32 B I
()	CFS DS Default: \$sc_\$cpu_DS_PassedPktCnt

Table 36 Telemetry Data –File Rate of Growth

	F
D	F
D	32 B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileState[].FileRate</i>

Table 37 Telemetry Data – File Size

	F
D	F
D	32 B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileState[].FileSize</i>

Table 38 Telemetry Data – File Open State

D	
D	16 B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileState[].OpenState</i>

Table 39 Telemetry Data - Filename Sequence Count

	C
D	C contains the sequence counter value that will be inserted into the filename of the next file to be created.
D	32 B I
()	<i>CFS DS Default: \$sc_\$cpu_DS_FileState[].FileSeq</i>

A.3 C

This section shows the DS Configuration Parameters provided as a default by the CFS DS application. Space is provided below for these values to be updated by the mission. If the mission has configured non-default values, they will be documented in the flight software RDL database.

While configuration parameters cannot be changed by the FOT, and are generally never changed after launch except by FSSE, the FOT needs to know the mission-specific values that have been incorporated into the software at the time the software was finalized and compiled.

(From an FSSE standpoint, configuration parameters are defined in the source code in ds_platform_cfg.h)

Table 40 Configuration Parameter - Application Pipe Depth

C	D A I E DE H
	<i>CFS DS Default: 256</i>
	A D
D	• D • D D
	CFE B MA I E DE H.

Table 41 Configuration Parameter - Application Pipe Name

C	D A I E AME
	<i>CFS DS Default: "DS_CMD_PIPE"</i>
	A
D	• D • FE B • "D .D CMD I E".
	() () (MA A I AME).

Table 42 Configuration Parameter - Destination File Table - Basename Buffer Size

C	D BA E AME B F I E
	<i>CFS DS Default: OS_MAX_PATH_LEN</i>
	D F - B B
D	• D F • -
	• • (MA A H LE).

Table 43 Configuration Parameter - Store DS Packet Processor State in CDS

C	D CD E ABLE A E
	<i>CFS DS Default: 1</i>
	E D C D
D	• D (CD) • A , A • C D , • D DEF E ABLE A E. • C D • D DEF E ABLE A E.

Table 44 Configuration Parameter - Destination File Table - Default Table Filename

C	D DEF DE FILE AME
	<i>CFS DS Default: "/cf/apps/ds_file_tbl.tbl"</i>
	D F - D F

D	D F .
	((MA) A H LE).

Table 45 Configuration Parameter - Default DS Packet Processor State

C	D DEF E ABLE A E
	<i>CFS DS Default: 1</i>
	D D
D	- D - D E ABLE

Table 46 Configuration Parameter - Packet Filter Table - Default Table Filename

C	D DEF FIL E FILE AME
	<i>CFS DS Default: "/cf/apps/ds_filter_tbl.tbl"</i>
	F - D F
D	F .
	((MA) A H LE).

Table 47 Configuration Parameter - Common Table File - Descriptor Text Buffer Size

C	D DE C I B F I E
	<i>CFS DS Default: 32</i>
	C F - D B

D	- D F D F

Table 48 Configuration Parameter - Destination File Table - Number of Files

C	D DE FILE C
	<i>CFS DS Default: 16</i>
	D F - F
D	- D D F - M -

Table 49 Configuration Parameter - Destination File Table - Logical Table Name

C	D DE I A I BL AME
	<i>CFS DS Default: "FILE_TBL"</i>
	D F - L
D	- FE D D F "D .FILE BL".
	CFE (BL MA AME LE G H.) FE

Table 50 Configuration Parameter - Destination File Table - Extension Buffer Size

C	D E E I B F I E
	<i>CFS DS Default: 8</i>
	D F - E B
D	• D F • -
	• • (MA A H L E).

Table 51 Configuration Parameter - Data Storage File - cFE File Header Description

C	D FILE HD DE C I I
	<i>CFS DS Default: "DS data storage file"</i>
	D F - F E F H D
D	D -
	C F E F HD DE C MA LE .

Table 52 Configuration Parameter - Data Storage File - cFE File Header Sub-Type

C	D FILE HD B E
	<i>CFS DS Default: 12345</i>
	D F - F E F H -
D	- D
	32 4,294,967,295.

Table 53 Configuration Parameter - File Header Type Selection

C	D FILE HEADE E
	<i>CFS DS Default: 1</i>
	F H
D	D F .
	: • 0 = -- • 1 = CFE -- FE • 2 = G M -- G M

Table 54 Configuration Parameter - Data Storage File - Minimum Age Limit

C	D FILE MI AGE LIM
	<i>CFS DS Default: 60</i>
	D F - M A L
D	() .
	C : A , .

Table 55 Configuration Parameter - Data Storage File - Minimum Size Limit

C	D FILE MI I E LIM
	<i>CFS DS Default: 1024</i>
	D F - M L
D	() .

	C : A ,
--	-------------------

Table 56 Configuration Parameter - Packet Filter Table - Logical Table Name

C	D FIL E BL AME
	<i>CFS DS Default: "FILTER_TBL"</i>
	F - L
D	• D F • FE . I • "D .FIL E BL".
	(CFE BL MA) AME LE G H. FE

Table 57 Configuration Parameter - Packet Filter Table - Filters Per Packet

C	D FIL E E ACKE
	<i>CFS DS Default: 4</i>
	F - F
D	• D F • ().
	D DE FILE C .

Table 58 Configuration Parameter - Make DS Tables Critical

C	D MAKE ABLE C I ICAL
	<i>CFS DS Default: 0</i>

	M D C
D	D ,
	.

Table 59 Configuration Parameter - Filename Sequence Count - Max Counter Value

C	D MA E E CE C
	<i>CFS DS Default: 99999999</i>
	F C - M C
D	.
	D E E CE DIGI ,

Table 60 Configuration Parameter - Mission Specific Version Number for DS Application

C	D MI I E
	<i>CFS DS Default: 0</i>
	M D A
D	- A : " . "
	M .

Table 61 Configuration Parameter - Move Files to Downlink Directory After Close Selection

C	D M E FILE
	<i>CFS DS Default: FALSE</i>

	M F D D A C
D	• D • •
	• E = D F • FAL E = D F

Table 62 Configuration Parameter - Packet Filter Table - Number of Packets

C	D ACKE I FIL E ABLE
	<i>CFS DS Default: 256</i>
	F -
D	• D F •
	• • CA I :

Table 63 Configuration Parameter - Destination File Table - Pathname Buffer Size

C	D A H AME B F I E
	<i>CFS DS Default: OS_MAX_PATH_LEN</i>
	D F - B
D	• D F •

	• • (MA A H LE).
--	--

Table 64 Configuration Parameter - Application per Packet Pipe Limit

C	D E ACKE I E LIM I
	<i>CFS DS Default: 50</i>
	A L
D	• • • D M ID ()
	D A I E DE H.

Table 65 Configuration Parameter - Housekeeping Request Frequency

C

	F C - D
D	• " " " " • ' .
	• " " - • D MA E E CE C ,

Table 67 Configuration Parameter - Data Storage File - Total Filename Size

C	D AL F AME B F I E
	<i>CFS DS Default: OS_MAX_PATH_LEN</i>
	D F - F
D	, , .
	• • (MA A H LE).)

D	I . : • C E C (\$sc_\$cpu_DS_CMDPC)* • C E C (\$sc_\$cpu_DS_CMDEC) * • I C (\$sc_\$cpu_DS_IgnoredPktCnt) * • F C (\$sc_\$cpu_DS_FilteredPktCnt) * • C (\$sc_\$cpu_DS_PassedPktCnt) * • F A C (\$sc_\$cpu_DS_FileWriteCnt) * • F A E C (\$sc_\$cpu_DS_FileWriteErrCnt) * • D L C (\$sc_\$cpu_DS_DestLoadCnt) * • F L C (\$sc_\$cpu_DS_FilterLoadCnt) * • D E C (\$sc_\$cpu_DS_DestPtrErrCnt) * • F E C (\$sc_\$cpu_DS_FilterPtrErrCnt) * <i>* or as defined by the mission</i>
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_ResetCtrs</i>
C	: • \$sc_\$cpu_DS_CMDPC (or as defined by the mission) - C • E ID 33
E C	(): • I
F E	E : • \$sc_\$cpu_DS_CMDEC (or as defined by the mission) - C • E ID 34 (H C)
C	

Table 70 Command 2 – Set Enable/Disable State For DS Application

C CF D C	2
C	E /D F D A
D	D . E /D D . D D D F D ,
C ()	<i>CFS DS Default:</i> <i>\$sc_\$cpu_DS_Enable (fixed command arg = enable)</i> <i>\$sc_\$cpu_DS_Disable (fixed command arg = disable)</i>
C ()	M D
C	: <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission) - C</i> - E ID 35
E C	(): - I - I /
F E	E : <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission) - C</i> - E ID 36 (A E /D)
C	F D .

Table 71 Command 3 – Set Destination File For Packet Filter Table Entry

C CF D C	3
---	---

C	D F F F E
D	D F F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetFilterFile</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • E ID 37
E C	(): • I • I F • I • I • F • C F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 38 (F F I)
C	

Table 72 Command 4 – Set Filter Type For Packet Filter Table Entry

C CF D C	4
C	F F F E
D	F F F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetFilterType</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • D F E CMD EID

E C	(): • I • I F • I • I • F • C F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 40 (F)
C	

Table 73 Command 5 – Set Algorithm Parameters For Packet Filter Table Entry

C CF D C	D E FIL E A M CC 5
C	A F F E
D	A F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetFilterParams</i>
C	: • \$ \$ D CMD C () - • E ID 41
E C	(): • I • I F • I • A A • F • C F
F E	E : • \$ \$ D CMDEC () - • E ID 42 (F)

C	
---	--

Table 74 Command 6 – Set Filename Type For Destination File Table Entry

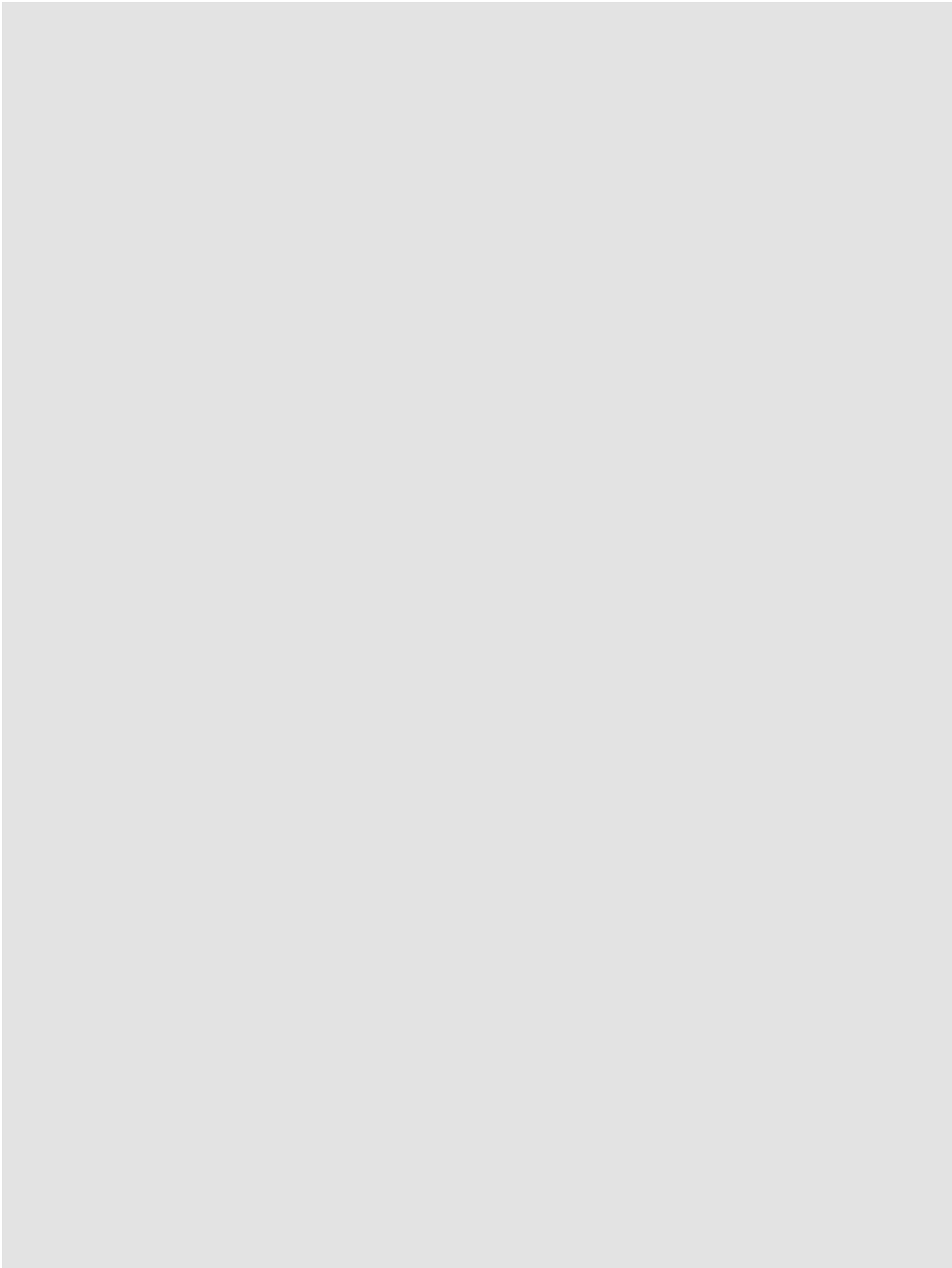
C CF D C	6
C	F F D F E
D	D F F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetFileType</i>
C	: • \$ \$ D CMD C () - • E ID 43
E C	(): • I • I D F • I • D F
F E	E : • \$ \$ D CMDEC () - • E ID 44 (F)
C	

Table 75 Command 7 – Set Enable/Disable State For Destination File Table Entry

C CF D C	7
C	E /D F D F E

D	D E /D F .
C ()	<i>CFS DS Default</i> • <i>\$sc_\$cpu_DS_EnableFile (fixed command arg = enable)</i> • <i>\$sc_\$cpu_DS_DisableFile (fixed command arg = disable)</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission) - C</i> • E ID 45
E C	(): • I • I D F • I • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission) - C</i> • E ID 46 (F E /D)
C	I F ID

Table 76 Command 8



E C	(): • I • I D F • I (,) • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission) -</i> C • E ID 50 (F B)
C	

Table 78 Command 10 – Set Extension For Destination File Table Entry

C CF D C	10
C	E F D F E
D	E D F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetExtension</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission) -</i> C • E ID 51
E C	(): • I • I D F • I (, ,) • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission) -</i> C • E ID 52 (F E)
C	

Table 79 Command 11 – Set Max File Size For Destination File Table Entry

C ^{CF D} _C	11
C	M F F D F E
D	D F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetMaxFileSize</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission) -</i> C • E ID 53
E C	(): • I • I D F • I • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission) -</i> C • E ID 54 (M F)
C	D / C . C F . I F .

Table 80 Command 12 – Set Max File Age For Destination File Table Entry

C ^{CF D} _C	12
C	M F A F D F E
D	D F .
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetMaxFileAge</i>

C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • E ID 55
E C	(): • I • I D F • I • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 56 (M F A)
C	/ C . D C . I F . .

Table 81 Command 13 – Set Sequence Count For Destination File Table Entry

C CF D C	D E DE C CC 13
C	C F D F E
D	D F . • F • F C . • - • H , FE ,
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_SetFileSeqCtr</i>

C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • E ID 57
E C	(): • I • I D F • I • D F
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 58 (C)
C	

Table 82 Command 14 – Close Destination File

C CF D C	D CL E FILE CC 14
C	C D F
D	D F
C ()	<i>CFS DS Default: \$sc_\$cpu_DS_CloseFile</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • E ID 59
E C	(): • I • I D F • D F

F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 60 (C D F)
C	

Table 83 Command 15 – Get File Info Packet

C CF D C	15
C	G F I
D	D F I .
C ()	<i>CFS DS Default: (N/A)</i>
C	: • <i>\$sc_\$cpu_DS_CMDPC (or as defined by the mission)</i> - C • D F I
E C	(): • I
F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 63 (G F I)
C	

Table 84 Command 16 – Add Message ID to Packet Filter Table

C CF D C	16
C	A M ID F
D	F M ID .

C ()	CFS DS Default: \$sc_\$cpu_DS_AddMID
C	: \$sc_\$cpu_DS_CMDPC (or as defined by the mission) - C - E ID 64
E C	(): - I - M ID () - F - M ID F
F E	E: \$sc_\$cpu_DS_CMDEC (or as defined by the mission) - C - E ID 65 (A M ID F)
C	

Table 85 Command 17 – Close All Destination Files

C CF D C	17
C	C A D F
D	- D F . - E:
C ()	CFS DS Default: \$sc_\$cpu_DS_CloseAll
C	: \$sc_\$cpu_DS_CMDPC (or as defined by the mission) - C - E ID 66 ()
E C	(): - I - D F

F E	E : • <i>\$sc_\$cpu_DS_CMDEC (or as defined by the mission)</i> - C • E ID 67
C	

A.5 E

This section shows cFE Software Bus event messages. Event messages are not mission specific. Generally, if supported by ground flight software systems (e.g., ASIST or ITOS), the Event ID numbers shown in the subsections below should appear in telemetry, no matter the mission.

A.5.1 E - C

The tables in this section show **critical** event messages for the DS application.

Table 86 Event ID 3 (Critical) - Application Termination

E D	3
E	'A E '
E	'A , = 0 %08 '
	C I ICAL
C	• FE D • () • (FE)

A.5.2 E - E

The tables in this section show **error** event messages for the DS application.

Table 87 Event ID 2 (Error) - Application Initialization

E D	2
E	'A I ()'
E	• ' E , = 0 %08 ' • ' , = 0 %08 ' • ' , = 0 %08 ' • ' 1H , = 0 %08 ' • ' D , = 0 %08 '

	E
C	- FE

Table 88 Event ID 6 (Error) - Critical Data Store Access

E D	6
E	'C D A ()'
E	'C D = 0 %08 '
	E
C	- D CD D F - CD CD D D

Table 89 Event ID 8 (Error) - Table Create

E D	8
E	' C ()
E	' F : E = 0 %08 ' ' D F : E = 0 %08 ' ' D F : F = '% ', E = 0 %08 ' ' F : F = '% ', E = 0 %08 '
	E
C	- A , D F - D F A , D . I D . I D D - FE - FE A I

Table 90 Event ID 11 (Error) - Destination File Validation

E	D	11
E		'D F ()
E		• 'D : ' , • 'D : = % , ' , • 'D : = % , ' , • 'D : = % , ' , • 'D : = % , = % ' , • 'D : = % , = % ' , • 'D : = % , = % ' , • 'D : = % , = % ' , • 'D : = % , = % ' ,
		E
C		• D D F . • () ,

Table 91 Event ID 13 (Error) – Packet Filter Table Validation

E	D	13
E		' F () '
E		• 'F : ' , • 'F : MID = 0 %04 , = % , = % , = % ' , • 'F : MID = 0 %04 , = % , = % , = % ' , • 'F : MID = 0 %04 , = % , = % , = % ' ,
		E

C	•
	•

Table 92 Event ID 14 (Error) - Filename Create

E D	14
E	'F C ()'
E	'FILE AME : = % , = '% ' , = '% ' , = '% ' , = '% ' "
	E
C	•
	•
	•

Table 93 Event ID 15 (Error) - File Create

E D	15
E	'F C ()'
E	'FILE C EA E : = % , = % , = '% ' "
	E

C	•
	(). H ,
	(. .
	).
	•
	, D F
	•
	-
	.

Table 94 Event ID 16 (Error) - File Write

E D	16
E	F ()'
E	FILE I E : = % , = % , = % , = '% ' "
	E
C	•
	•
	().
	•
	, D F
	•
	-
	.

Table 95 Event ID 21 (Error) - Invalid Packet

E D	21
E	'l ()'
E	'l : MID = 0 %04 , CC = % ' '
	E
C	•
	D (CC). G
	" "
	-
	•
	M ID (MID)
	.

Table 96 Event ID 22 (Error) - Housekeeping Request

E D	22
E	'H ()'
E	'l HK : = % , = % '
	E
C	• HK •

Table 97 Event ID 32 (Error) - No-op

E D	32
E	' - ()'
E	'l : = % , = % '
	E
C	.

Table 98 Event ID 34 (Error) - Reset Housekeeping Counters

E D	34
E	' H C ()'
E	'l E E : = % , = % '
	E
C	E E .

E	• 'I A A E : = % , = % ' • 'I A A E : = % '
	E
C	• / . / . • .

Table 100 Event ID 38 (Error) - Set Filter File Index

E D	38
E	' F F I ()'
E	• 'I FIL E FILE : = % , = % ' • 'I FIL E FILE : ID = 0 %04 ' • 'I FIL E FILE : = % ' • 'I FIL E FILE : = % ' • 'I FIL E FILE : , ' • 'I FIL E FILE : M ID 0 %04 , '
	E
C	• F F , M ID • F M ID • F . • .

Table 101 Event ID 40 (Error) - Set Filter Type

E D	40
E	' F ()'
E	• 'I FIL E E : = % , = % ' • 'I FIL E E : ID = 0 %04 ' • 'I FIL E E : = % ' • 'I FIL E E : = % ' • 'I FIL E E : , ' • 'I FIL E E : M ID 0 %04 , '

	E
C	• F , M ID, F • F M ID • F

Table 102 Event ID 42 (Error) - Set Filter Parameters

E D	42
E	' F ()'
E	• 'I FIL E A M : = % , = % ' • 'I FIL E A M : ID = 0 %04 ' • 'I FIL E A M : = % ' • 'I FIL E A M : = % , = % , = % ' • 'I FIL E A M : ' • 'I FIL E A M : M ID 0 %04
	E 3
C	• F , M ID, (, ,). • F M ID • F

Table 103 Event ID 44 (Error) - Set Filename Type

E D	44
E	' F ()'
E	• 'I DE E : = % , = % ' • 'I DE E : = % ' • 'I DE E : = % ' • 'I DE E :

	E
C	• D F , D F • D F •

Table 104 Event ID 46 (Error) - Set File Enable/Disable State

E D	46
E	' F E /D ()'
E	• 'I DE A E : = % , = % ' • 'I DE A E : = % ' • I DE A E : = % ' • I DE A E :
	E
C	• / D F , - D F • D F •

Table 105 Event ID 48 (Error) - Set Filename Path

E D	48
E	' F ()'
E	• 'I DE A H : = % , = % ' • 'I DE A H : = % ' • 'I DE A H : • 'I DE A H :
	E

C	•
	D F , D F
	• D F

Table 106 Event ID 50 (Error) - Set File Basename

E D	50
E	' F B () '
E	• 'I DE BA E : = % , = % ' • 'I DE BA E : = % ' • 'I DE BA E : ' • 'I DE BA E :
	E
C	•
	D F , D F
	• D F

Table 107 Event ID 52 (Error) - Set Filename Extension

E D	52
E	' F E () '
E	• 'I DE E : = % , = % ' • 'I DE E : = % ' • 'I DE E : ' • 'I DE E :
	E

C	•
	D F , D F
	• D F

Table 108 Event ID 54 (Error) - Set Max File Size

E D	54
E	' M F ()'
E	• 'I DE I E : = % , = % ' • 'I DE I E : = % ' • 'I DE I E : = % ' • 'I DE I E : '
	E
C	•
	D F , D F
	• D F

Table 109 Event ID 55 (Error) - Set Max File Age

E D	55
E	' M F A ()'
E	'DE AGE : = % , = % '
	DEB G
C	•
	D F , D F
	• D F ,

Table 110 Event ID 56 (Error) - Set Max File Age

E D	56
-----	----

E	' M F A ()'
E	• 'I DE AGE : = % , = % ' • 'I DE AGE : = % ' • 'I DE AGE : = % ' • 'I DE AGE : '
	E
C	• D F , D F • D F •

Table 111 Event ID 58 (Error) - Set Sequence Count

E D	58
E	' C ()'
E	• 'I DE C : = % , = % ' • 'I DE C : = % ' • 'I DE C : = % ' • 'I DE C : '
	E
C	• D F , D F • D F •

Table 112 Event ID 60 (Error) - Close Destination File

E D	60
E	'C D F ()'
E	• 'I DE CL E : = % , = % ' • 'I DE CL E : = % '

	E
C	D F . D F .

Table 113 Event ID 61 (Error) - File Move

E D	61
E	'F M ()
E	'FILE M E : = '% ', = '% ' 'FILE M E : = '% ', = '% ', = % '
	E
C	D ,

Table 114 Event ID 63 (Error) - Get File Info

E D	63
E	'G F I ()'
E	'I GE FILE I F : = % , = % '
	E
C	

Table 115 Event ID 65 (Error) - Add Message ID to Packet Filter Table

E D	65
E	'A M ID F ()'

E	• 'I ADD MID : = % , = % ' • 'I ADD MID : MID = 0 %4 ' • 'I ADD MID : ' • 'I ADD MID : MID = 0 %4 = % ' • 'I ADD MID : '
	E
C	• M ID F . • M ID. • F • M ID • M ID. • .

Table 116 Event ID 67 (Error) - Close All Destination Files

E D	67
E	'C A D F ()'
E	'I DE CL E ALL : = % , = % '
	E
C	• D F . • .

A.5.3 E -

The tables in this section show **information** type event messages for the DS application.

Table 117 Event ID 1 (Informational) - Application Initialization Event

E D	1
E	'A I E '
E	'A , % .% .% .% , 0 %08 '

	I F MA I
C	• %d D A ' M • %d A ' M • %d A ' M • %d A ' M

Table 118 Event ID 10 (Informational) - Destination File Table Validation Results

E D	10
E	'D F '
E	'D = % ' : = % , = % , = % ,
	I F MA I
C	• FE M D F • () (K), () , () . + +

Table 119 Event ID 12 (Informational) – Packet Filter Table Validation Results

E D	12
E	' F '
E	'F : = % , = % , = % , = % '
	I F MA I

C	•	FE	M	F
	•	(	)	(K),
		(	)	,
		(	)	.
	+	+		.

Table 120 Event ID 31 (Informational) - No-op

E	D	31
E		' - ()'
E		' , % .% .% .% '
		I F MA I
C	•	D
	•	% M
	•	% M
	•	%
	•	% M

A.5.4 E - D

The tables in this section show **debug** event messages for the DS application.

Table 121 Event ID 7 (Debug)

E	D	7
E		' F CD '
E		'D F CD '
		'F CD '
		DEB G

C	• I D , D • CD
----------	---

Table 122 Event ID 33 (Debug)

E D	33
E	' H C '
E	' '
	DEB G
C	E E

Table 123 Event ID 35 (Debug)

E D	35
E	' A E /D '
E	'A A E : = % '
	DEB G
C	• / • D • /

Table 124 Event ID 37 (Debug)

E D	37
E	' F F I '
E	'FIL E FILE : MID = 0 % , = % , = % , = % '
	DEB G

C	• F M ID. ,
	• E ,
	• F D F .
	• F .
	• M ID, F F D F

Table 125 Event ID 39 (Debug)

E D	39
E	' F '
E	'FIL E E : MID = 0 %04 , = % , = % , = % '
	DEB G
C	• F M ID. ,
	• E ,
	• F .
	• M ID, F F
	• ,

Table 126 Event ID 41 (Debug)

E D	41
E	' F '
E	'FIL E A M : MID = 0 %04 , = % , = % , = % , = % '
	DEB G

C	• F M ID.
	• E ,
	• ,
	• F . M ID, F (, ,).

Table 127 Event ID 43 (Debug)

E D	43
E	' F ()'
E	'DE E : = % , = % '
	DEB G
C	• D F .
	• D F ,

Table 128 Event ID 45 (Debug)

E D	45
E	' F E /D '
E	'DE A E : = % , = % '
	DEB G
C	• / D F .
	• / D F ,

Table 129 Event ID 47 (Debug)

E D	47
E	' F '
E	'DE A H : = % , = '% "

	DEB G
C	• D F . • D F ,

Table 130 Event ID 49 (Debug)

E D	49
E	' F B ()'
E	'DE BA E : = % , = '% "
	DEB G
C	• D F . • D F ,

Table 131 Event ID 51 (Debug)

E D	51
E	' F E '
E	'DE E : = % , = '% "
	DEB G
C	• D F . • D F ,

Table 132 Event ID 53 (Debug)

E D	53
E	' M F '
E	'DE I E : = % , = % '

	DEB G
C	• D F . • D F ,

Table 133 Event ID 57 (Debug)

E D	57
E	' C ()'
E	'DE C : = % , = % '
	DEB G
C	• D F . • D F ,

Table 134 Event ID 59 (Debug)

E D	59
E	'C D F '
E	'DE CL E : = % '
	DEB G
C	• D F . • D F ,

Table 135 Event ID 62 (Debug)

E D	62
E	'G F I '
E	'GE FILE I F '

	DEB G
C	• •

Table 136 Event ID 64 (Debug)

E D	64
E	'A M ID F '
E	'ADD MID : MID = 0 %04 , = % '
	DEB G
C	• M ID F • F M ID. M ID

Table 137 Event ID 66 (Debug)

E D	66
E	'C A D F ()'
E	'DE CL E ALL '
	DEB G
C	• D F •

Appendix B **D**

B.1 - **C**

- This document presents selected *information in this dark orange font* that should be removed when tailoring this document for a mission.
- Command and Telemetry mnemonics are mission-specific. This document as delivered has “suggested” names that may or may not be used by the mission when the MOT creates the ground system database. In particular, the suggested names start with \$sc_\$cpu_DS which indicate a global setting for spacecraft, processor selection, and the DS subsystem. This has meaning if the mission has multiple spacecraft, each with a copy of cFE/CFS apps being executed, and/or multiple processors per spacecraft, each with a copy of cFE/CFS apps being executed. Most missions have neither and they do not prepend a \$sc_\$cpu_ selection to the front of the command name. However it is common for missions to differentiate the spacecraft subsystem commands from instrument commands by prepending a couple of characters (e.g. pw for power system) to all the command and telemetry names for that subsystem.
- The nomenclature of command and telemetry mnemonics is highly mission-specific, so this document does not attempt to include the actual command and telemetry database names in advance. For example, for MMS the telemetry mnemonics are defined in an RDL file for ASIST, and they will not exactly match the mnemonics in this Guide.

B.2 **D**

This section is for anyone tailoring this Guide for a specific mission.

When tailoring this document for a particular mission:

- *Remove or replace text appearing in this dark orange font, with mission-specific values.*
- Review figures to be sure there are no conflicts with mission configurations. Edit figures with MS Visio or MS PowerPoint if necessary.
- Replace CFS defaults with mission defined values in Appendix A.
- Regenerate the table of contents (TOC); add Appendix page prefixes to TOC as needed. Table of Contents, Figures, and Tables can be updated automatically, but the letter prefix for Appendix pages must be added manually after update. To update all figure and table references in the document, when using the PC version of Word, select all, then choose F9.
- When complete, remove Appendix B.

This Guide is formatted using Microsoft Word styles. When adding new sections to the Guide, assign paragraphs to the styles shown in the table below. Center tables and figures horizontally on

the page. Use 15% grayscale in new table headings. For bullets in tables, assign the style shown below to set vertical spacing, then assign bullets using the bullet icons menu.

Table 138 Internal Document Styles

C	H 1 6	L
F	L B 1 .	L
	L B 2	L
	L E 0	L
	C	/A
A	B	F
& F		
F	H	C
A	C	L
F	C F	C
	C	C
B	C +	L